

Hierarchical information mediates time-course of access to items in linguistic memory

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Quick Summary

Question: How does hierarchical syntactic information shape memory processes in real-time dependency formation?

→ **Hypothesis 1:** via hierarchically informed cues distinguishing hierarchically available representations at retrieval (Kush, 2013)

→ **Hypothesis 2:** via organizing representations into distinct memory states during encoding (McElree et al., 2003; Wagers & McElree, 2022)

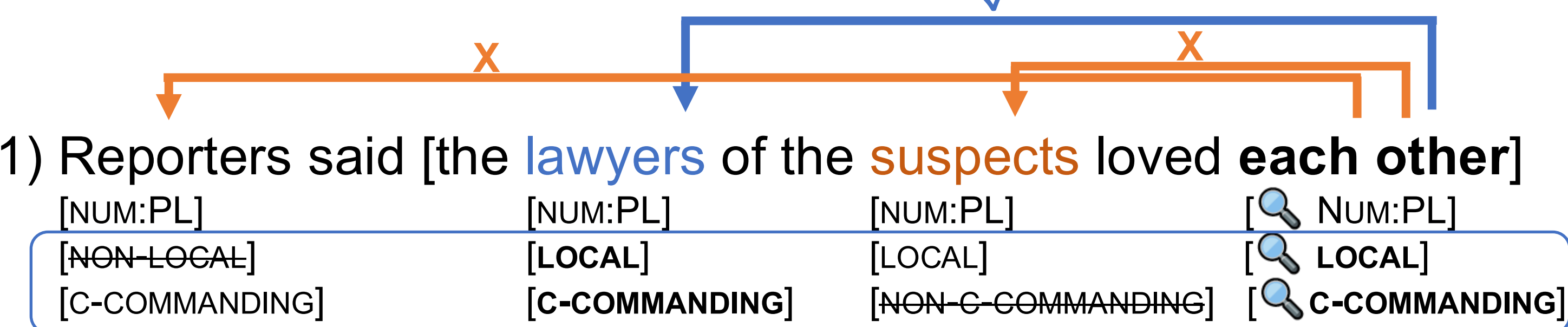
Finding: Hierarchically available targets are accessed earlier than unavailable distractors in processing a local anaphor.

Implication: A functional divide within linguistic memory shaped by hierarchical syntactic structure, with hierarchically available representations in a focal, privileged state

The Study

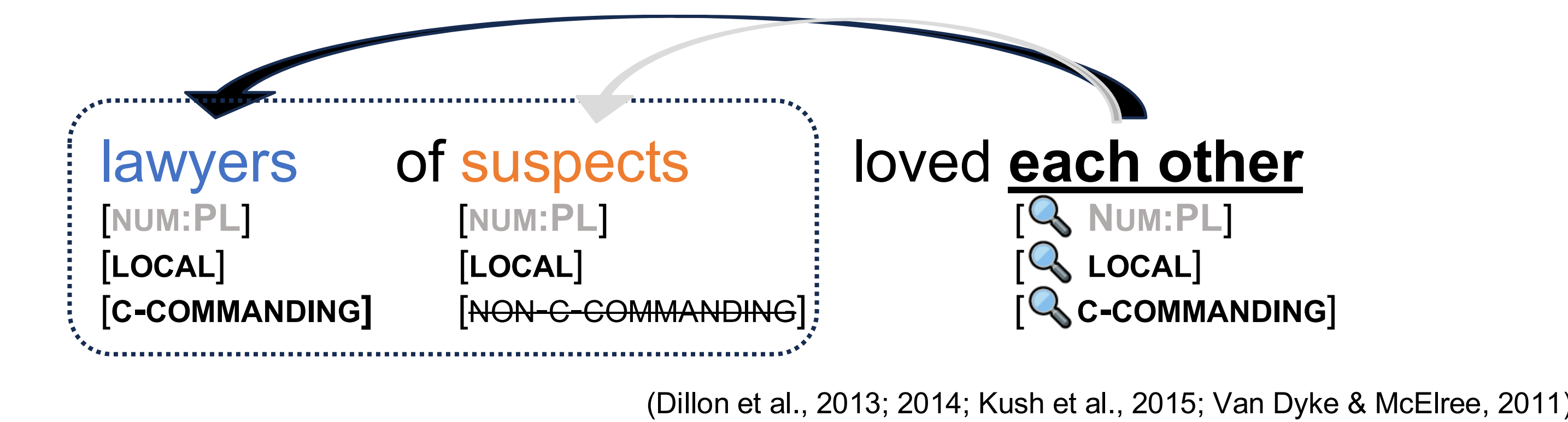
➤ Empirical evidence for rapid use of **hierarchical syntactic relations** in forming anaphor-antecedent dependencies

(Bakay et al., 2026; Dillon et al., 2013; 2014; Kush et al., 2015; Moulton & Han, 2018; Sturt, 2003)

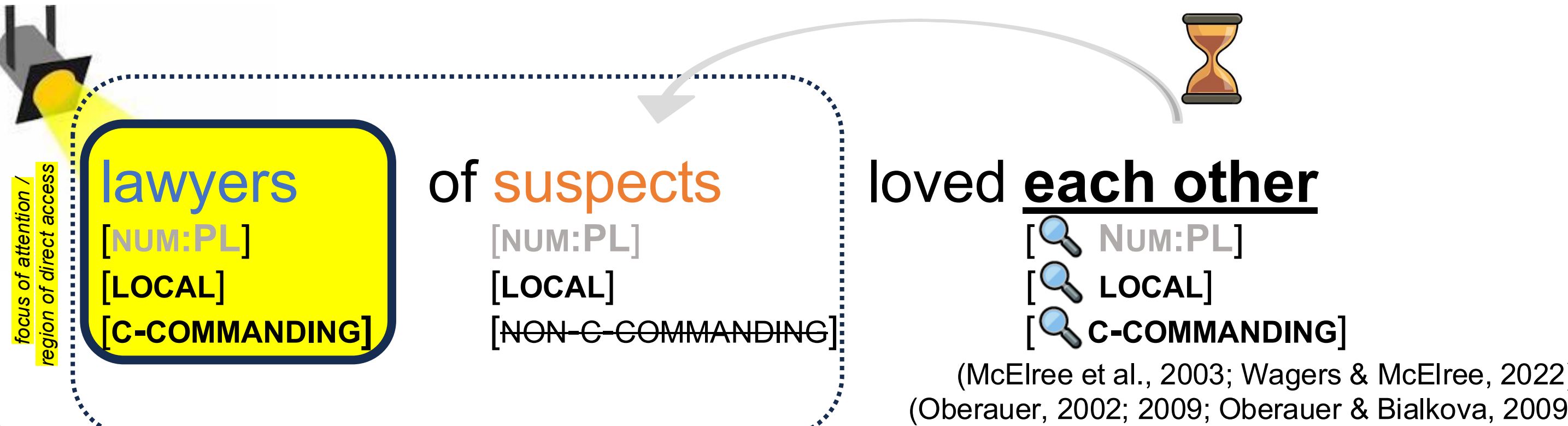


? How does **hierarchical structural information** inform memory access to linguistic representations?

Hypothesis 1: Hierarchically guided retrieval



Hypothesis 2: Privileged hierarchical representations



Divergence points in gaze proportions

($N_{\text{participant}} = 152$)

- **Turkish** stimuli with a preverbal, local reciprocal anaphor
- Hierarchically available (*c-commanding*, *co-argument*) target **subject** and **IO** vs. unavailable **distractor**
- **IO** and **dist** kept similar in number, clause & linear position

2) a. Ayşe [subj-PL-GEN IO-PL-DAT Adv birbirleri V] V
b. Ayşe [subj-PL-GEN dist-PL-DAT P Adv birbirleri V] V

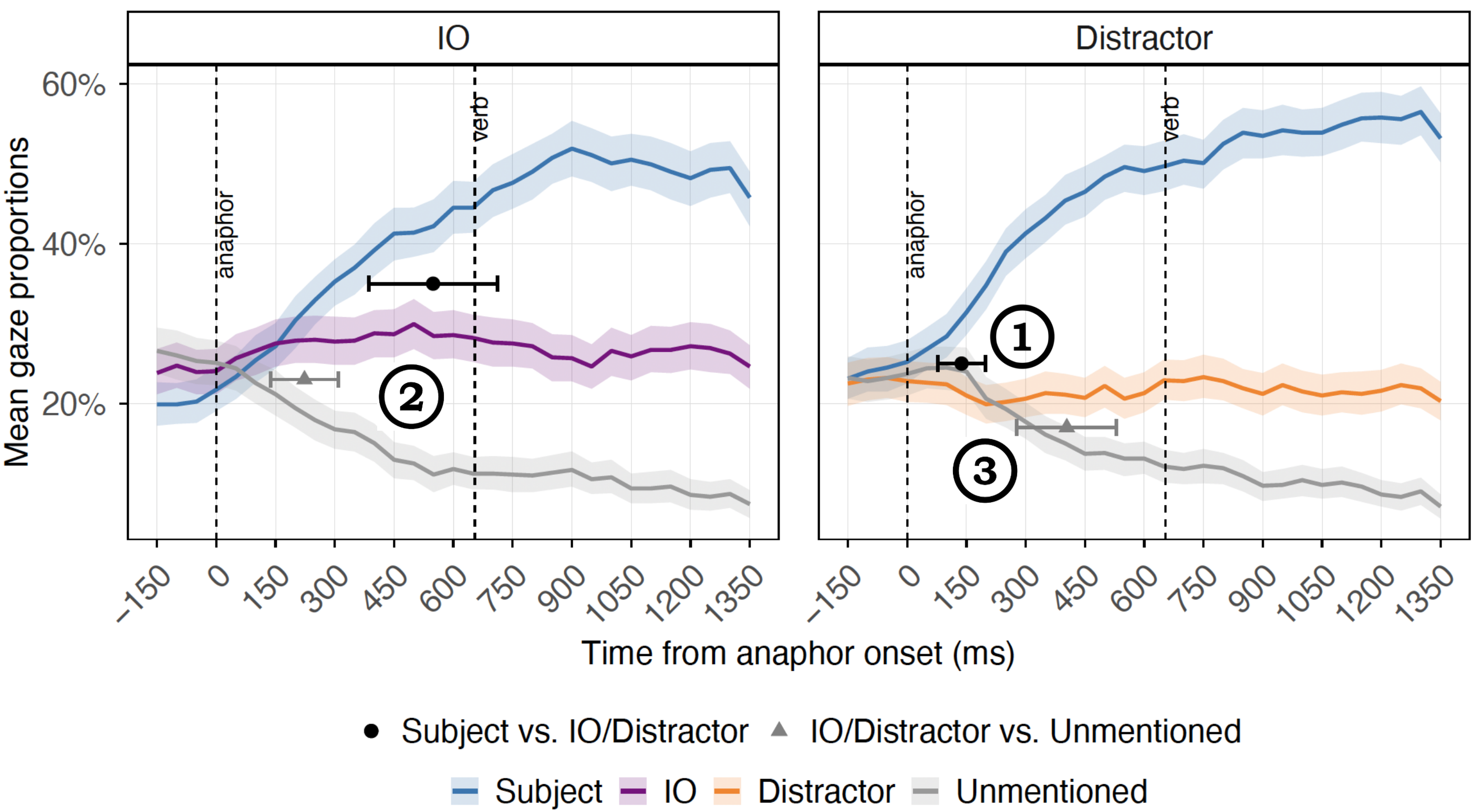


Example visual display



Scan for stimuli

➤ Bayesian estimation of divergence onsets in proportions of looks to referents via bootstrapping (Fernandez et al., 2025; Stone et al., 2020)

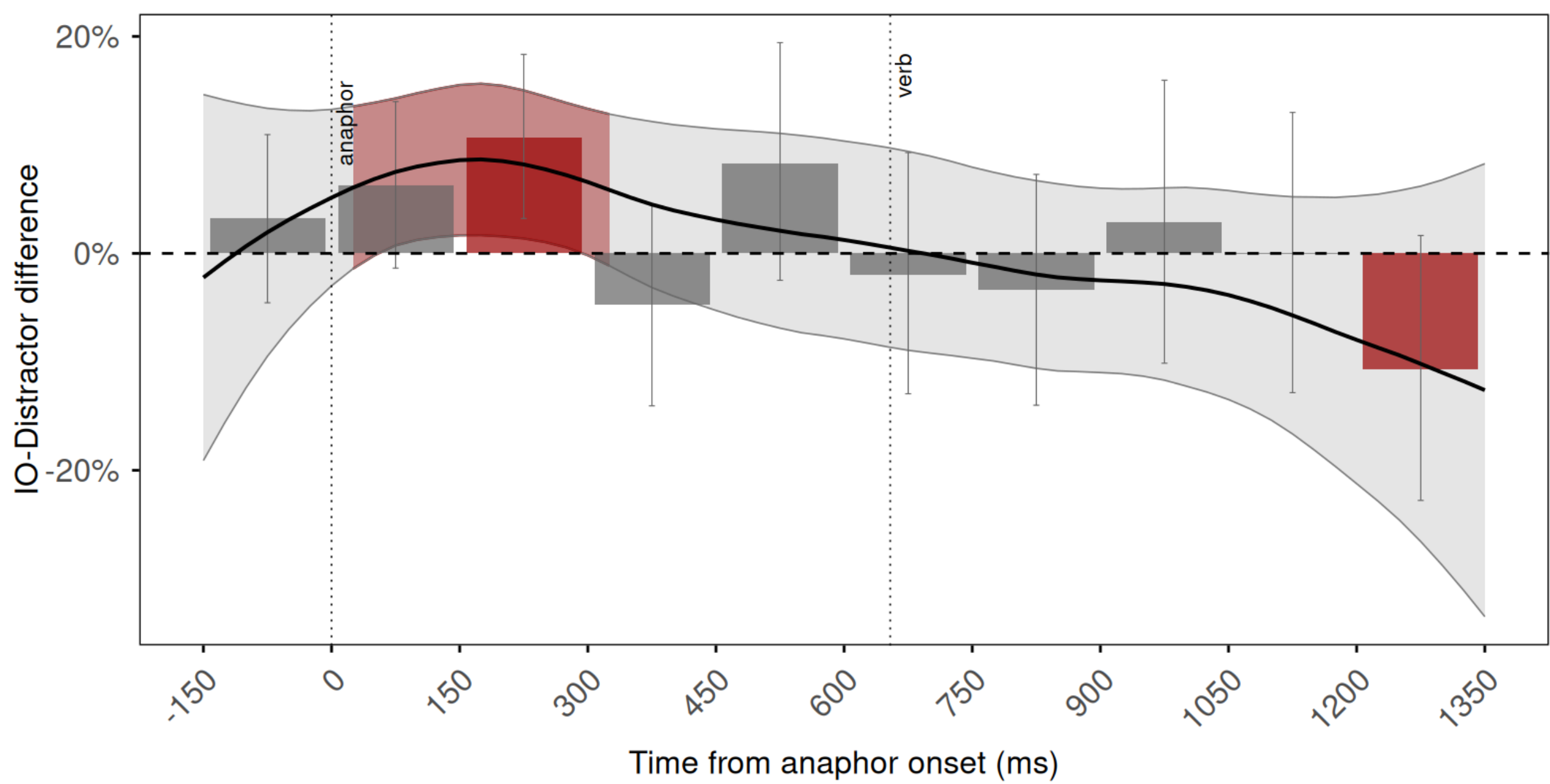


In trials with a **subj** response:

- ① **subj** vs. **dist** ~138ms [77, 198]
✓ rapid use of hierarchical information to distinguish critical referents within the anaphor window
- ② **unmentioned** vs. **IO** ~223ms [138, 309]
- ③ **unmentioned** vs. **dist** ~404ms [278, 530]
difference between the two ~181ms [28, 333], $P(\hat{\beta} > 0) = 0.99$, $BF_{10} = 2.2$
✓ delay in reactivating dist relative to **IO**

Time-course of new fixations

- Bayesian analyses on the time-course of *new* fixations launched within the anaphor and verb windows (Kingston et al., 2016)
 $P(\text{new fixations to IO vs. dist} \mid \text{all new fixations})$
→ Overall slightly more new fixations towards **IO** than **dist**
 $\hat{\beta} = -0.13$, $\text{Err} = 0.08$ [-0.29, 0.03], $P(\hat{\beta} > 0) = 0.06$, $BF_{10} = 0.28$
- More new fixations towards **IO** in **early processing**
150-300ms time-bin: 11% [3, 18] $P(\hat{\beta} > 0) = 0.99$, $BF_{10} = 14.2$
25-325ms in difference spline: 6-9% $P(\hat{\beta} > 0) \geq 0.95$
- More new fixations towards **dist** in **late processing**
1200-1350ms time-bin: -11% [-23, 2] $P(\hat{\beta} > 0) = 0.04$, $BF_{10} = 1.4$
- ✓ Earlier access to IO relative to **dist**



Privileged hierarchical representations

- Hierarchical syntactic information organizes representations across active vs. passive memory states → **Hypothesis 2**
- Transient associations link content (linguistic representations) with context (hierarchical structure) in memory; (Keshev et al., 2025)
- Hierarchically prominent representations are kept in temporarily active, privileged memory states via their structural binding, as in: (McElree, 2006; Oberauer, 2002; 2009)
focus of attention or region of direct access;
- Representations within this privileged state (**subj**, **IO**) are readily available and accessed faster;
- Those that aren't (**dist**) lack this privileged state and are accessed slowly via an explicit retrieval. (Lewis & Vasishth, 2005)



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